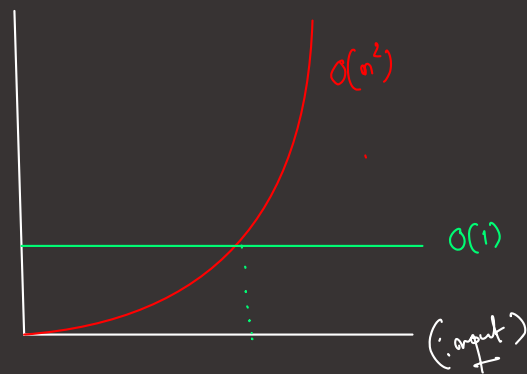


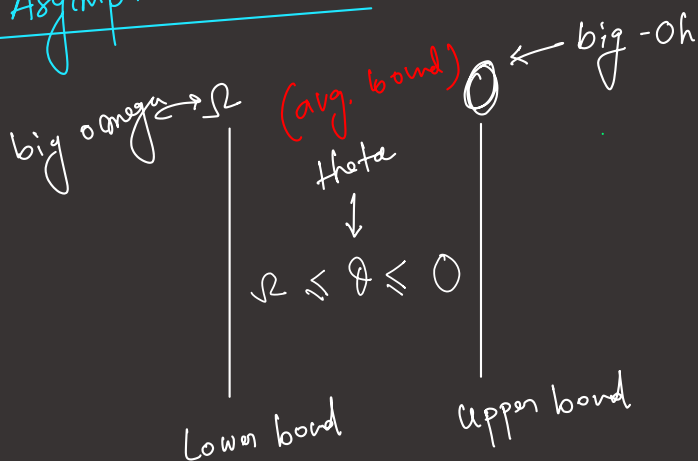
$$f(n) = an^2 + n \log n + c \rightarrow O(n^2)$$

$$f(n) = 2n^2 \log n + n \log n + n^2 + 5$$

$$\rightarrow O(n^2 \log n)$$



Asymptotic Notation



$$1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < \dots < 2^n < 3^n < \dots < n^n$$

Big-Oh

The function $f(n) = O(g(n))$ iff $\exists$ +ve constant c and n_0 ,
such that, $f(n) \leq c \cdot g(n) \quad \forall n \geq n_0$.

$$f(n) = 2n + 3 \leq \frac{10n}{c} \quad \forall n \geq n_0$$

$\uparrow$
 $g(n)$

$$2n + 3 < 10n^2 \quad \forall n \geq 1$$

$$2n + 3 < 2n + 3n = 5n$$

$$2n + 3 < 5n \quad \forall n \geq 1$$

$$f(n) = O(n)$$

$$= O(n^2)$$

$$= O(n \log n)$$

$$= O(\log n) \quad \text{X}$$

Big-Omega

The function $f(n) = \Omega(g(n))$ iff $\exists$ +ve constant c and n_0 ,
such that, $f(n) \geq c \cdot g(n) \quad \forall n \geq n_0$.

$$f(n) = 2n + 3 \geq 1 \cdot n \quad \forall n \geq 1$$

$\swarrow \quad \nwarrow$
 $c \quad g(n)$

$$\begin{aligned} f(n) = 2n + 3 &= \Omega(n) \\ &= \Omega(\log n) \\ &= \Omega(\sqrt{n}) \\ &= \Omega(1) \\ &= \Omega(n^2) \times \end{aligned}$$

Theta Notation

The function $f(n) = \Theta(g(n))$ iff $\exists$ +ve constant c_1, c_2 and n_0
such that, $c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n) \quad \forall n \geq n_0$

$$f(n) = 2n + 3$$
$$c_1 \cdot \underbrace{1}_{g(n)} \cdot \underbrace{n}_{g(n)} \leq 2n + 3 \leq \underbrace{5}_{c_2} \cdot \underbrace{n}_{g(n)} \quad \forall n \geq 1$$

$\therefore f(n) = \Theta(n)$
 $= \Theta(n^2) \times$

Ex $f(n) = n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-2)(n-1)n \leq \underbrace{n \cdot n \cdot n \cdot \dots \cdot n \cdot n \cdot n}_{n \text{ times}} = n^n$

$$1 \leq n! \leq n^n \quad \forall \underline{n \geq 1}$$

$$n! = O(n^n) \\ = \Omega(1)$$

$\Theta(?) \rightarrow$ not possible to say

Q $f(n) = \log(n!)$ $1 \leq \log(n!) \leq n \log n \quad \forall n \geq 1$

Linear Search

key = 10

key = 13, 20

1	5	10	6	9	4	3	1	2	13
0	1	2	3	4	5	6	7	8	9

Best Case $B(n)$
Time Complexity = $O(1)$

Worst Case $W(n)$
Time Complexity = $O(n)$

Average Case

Time Compⁿ =

$$\frac{\text{all possible case time}}{\text{no. of cases}}$$

$$= \frac{1+2+3+\dots+n}{n} = \frac{n(n+1)}{2 \cdot n}$$

$$= \frac{n+1}{2} = O(n)$$

Avg. Case $A(n)$
T.C = $O(n)$

$$B(n) = O(1) = \Omega(1)$$

$$= O(1)$$

$$= \Omega(1)$$

$$= \Theta(1)$$

$$W(n) = n$$

$$= O(n)$$

$$= \Omega(n)$$

$$= \Theta(n)$$


```

for (i=0; i < n; i++) {
    for (j=i+1; j < n; j++) {
        start;
    }
}

```

$i=0, j=1 \text{ to } (n-1), \text{start} \rightarrow (n-1)$
 $i=1, j=2 \text{ to } (n-1), \text{start} \rightarrow (n-2)$
 $i=2, j=3 \text{ to } (n-1), \text{start} \rightarrow (n-3)$
 $\vdots$
 $i=(n-2), j=n-1 \text{ to } n-1, \text{start} \rightarrow 1$
 $i=(n-1), j=n \times \text{start} \rightarrow 0$
 $i=n \times$

$$\therefore \text{time } f_n^n: 1 + 2 + \dots + (n-1)$$

$$= \frac{n(n-1)}{2}$$

$$\therefore \text{C.C} = O(n^2)$$

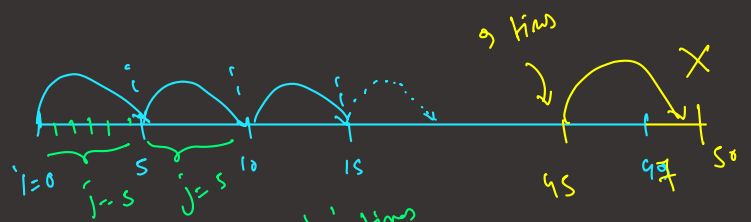
assume $k < n$

```

for (i=0; i < n; i=i+k) {
    for (j=1; j <= k; j++) {
        start
    }
}

```

$n=48$
 $k=5$



for each sub-loop inner loop runs 'k' times

$\therefore \text{time } f_n^n:$

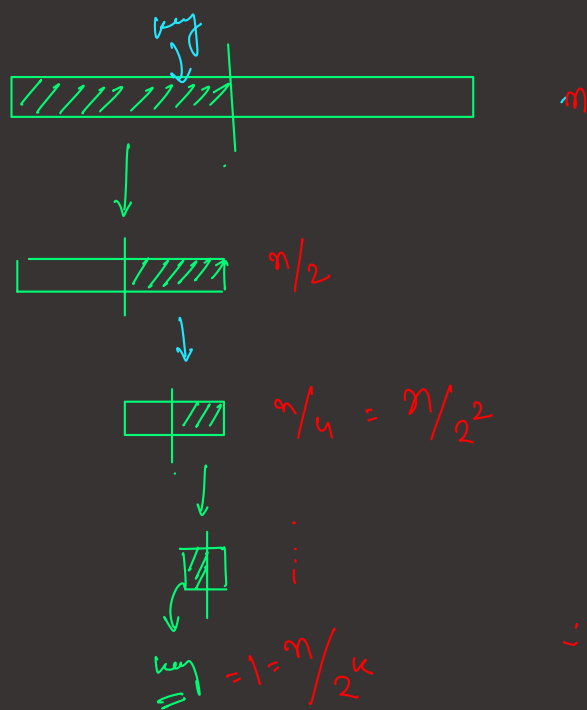
$$\frac{n}{k} \cdot k = n$$

$$\therefore O(n)$$

How many times this outer loop is running?

$$\left\lfloor \frac{n}{k} \right\rfloor \text{ times?}$$

day-18
Binary Search



loop for k times

$$\therefore \frac{n}{2^k} = 1$$

$$\Rightarrow 2^k = n$$

$$\Rightarrow k = \log n$$

$$\therefore O(\log n)$$

Recursive Equation

$$T(n) = k + T(n/2)$$

$f(n)$
 $f(\text{ans}, i, j)$
 k mid // steps
 $n/2$ $2f$ or RH
 $f(n/2)$

Seq

$$T(n) = k + T(n/2)$$

$$T(n/2) = k + T(n/4)$$

$$T(n/4) = k + T(n/8)$$

$\vdots$

$$P() = k + 0$$

i times

$$\frac{n}{2^i} = 1$$

$$i = \log n$$

$$T(n) = \begin{cases} k + T(n/2) & n > 1 \\ 0, 0/w & \text{otherwise} \end{cases}$$

$$\therefore T(n) = k \log n$$

$$= \underline{\underline{O(\log n)}}$$

Merge Sort

$$f(n) = f(n/2) + f(n/2) + \text{merge}()$$

$$\therefore T(n) = T(n/2) + T(n/2) + O(n)$$

$$\Rightarrow T(n) = 2T(n/2) + nk$$

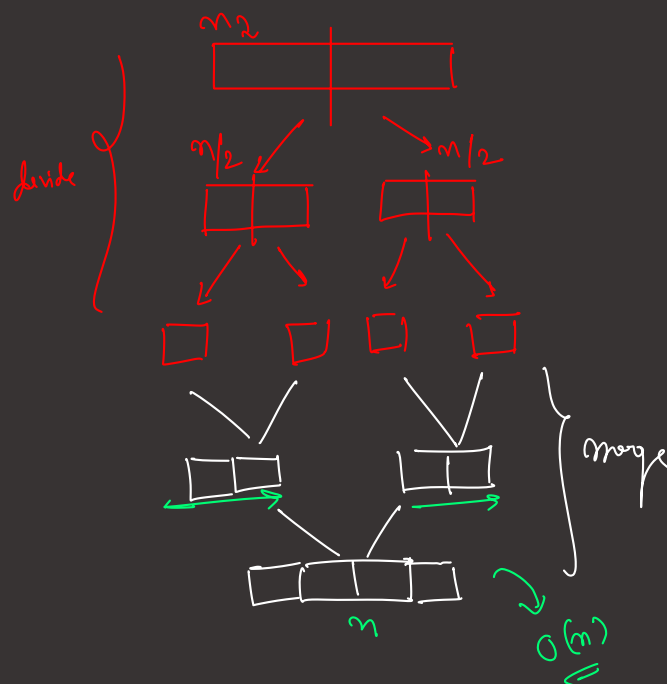
$$\Rightarrow 2T(n/2) = 4T(n/4) + nk$$

$$\Rightarrow 4T(n/4) = 8T(n/8) + nk$$

$$\Rightarrow 8T(n/8) = 16T(n/16) + nk$$

$\vdots$

$$\Rightarrow nT(1) = 0 + nk$$



$\therefore \text{times} = \log(n)$

$$\therefore T(n) = nk \log n = k(n \log n)$$

$$= \underline{O(n \log n)}$$